QwenPI VLA Training Optimization

NVIDIA DevTech

Results at a Glance

starVLA QwenPI: Qwen3.5-VL 4B + flow-matching action head, ZeRO-2 full fine-tune

3.2×

Faster training step

800 → 249 ms/step (bs 8)

257

Samples / node / s

up from 80 at hand-off

23.1%

Model FLOPs utilization

up from 7.2% (bf16 peak)

340

Samples / node / s

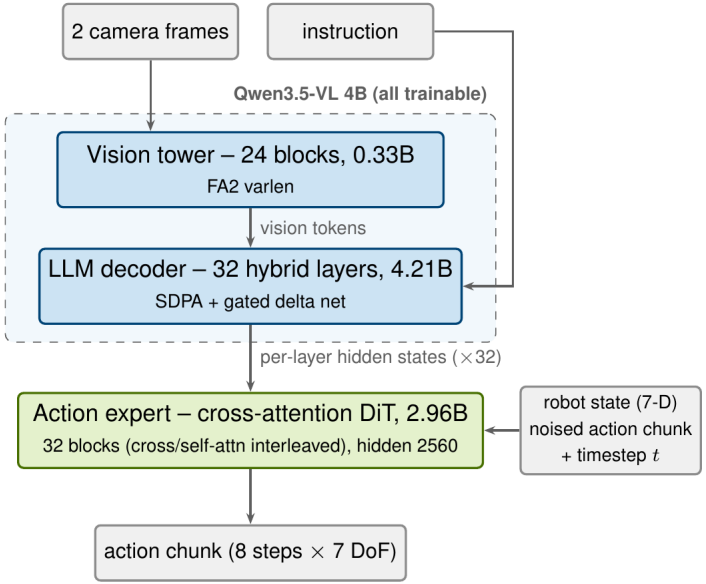
at bs 24 — 1.7× the 200 target

Method: every optimization was profiled with Nsight Systems and validated with a same-node A/B run plus loss-trajectory checks — no stacked guesses, measured deltas only.

Measurement: 8×H200 (NVLink), 60 steps, steady-state p50 of steps 21–44, per-GPU batch 8 unless noted.

Model & Workload

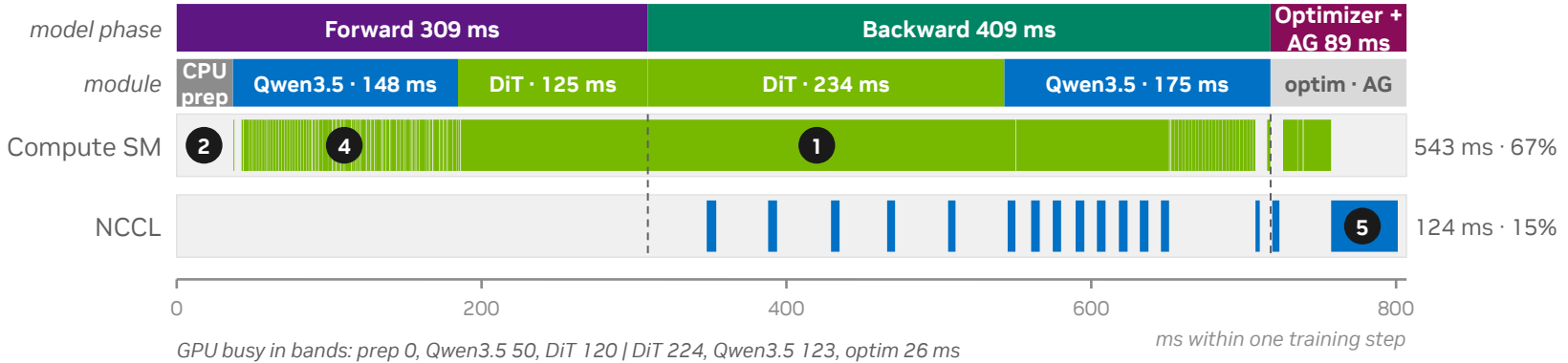
7.5B trainable parameters: Qwen3.5-VL 4B backbone + 3B flow-matching action expert



Training configuration	
Parameters	7.52B, all trainable (bf16)
Parallelism	DeepSpeed ZeRO-2, DP=8, 1 node
ZeRO-2 tuning	bucket 1.5e9, reduce_scatter off
Optimizer	AdamW, fp32 master (sharded)
Batch	8 / GPU → 64 global (up to 24/GPU)
Sequence	text fixed at 192 (left-pad)
Data	LIBERO manipulation, 2 views/sample
Hardware	8×H200 SXM, full NVLink
Software	torch 2.7.1 / NCCL 2.26.2 / HF 5.3 / FA2.8

Baseline Diagnosis: Where Did 800 ms Go?

One baseline step on the GPU-execution clock (807 ms, bs 8/GPU) — five problems, five fixes

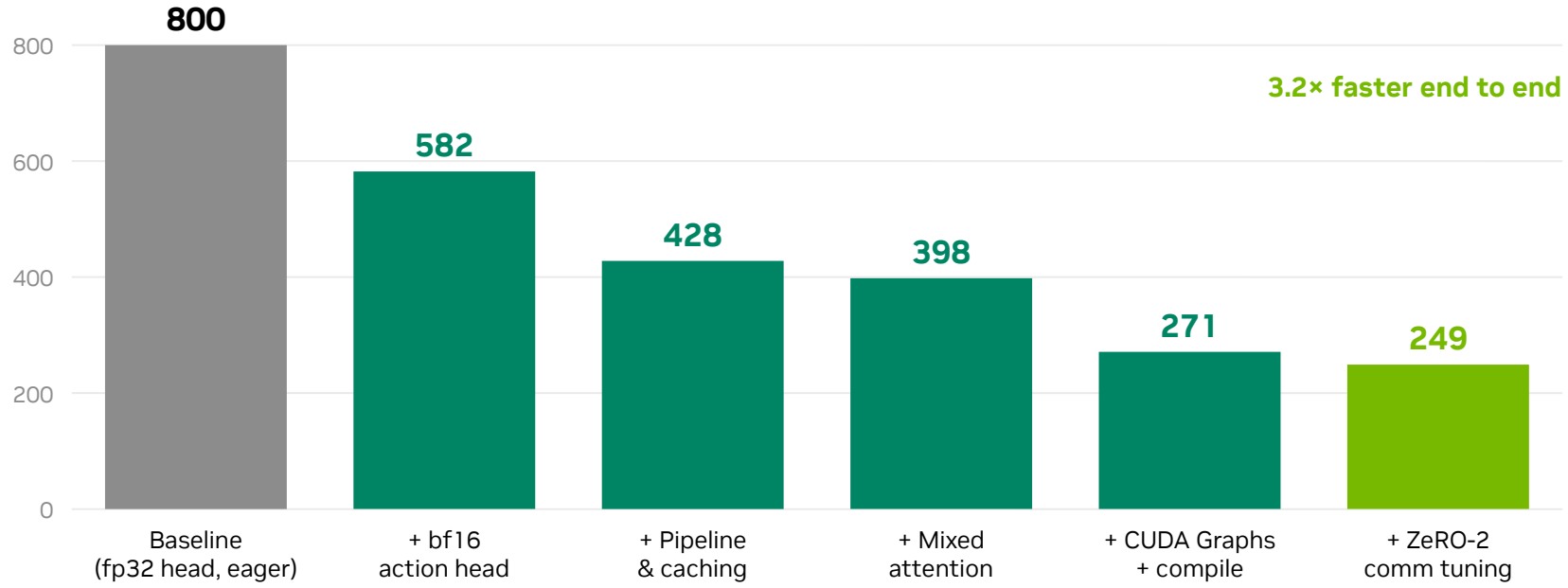


GPU busy in bands: prep 0, Qwen3.5 50, DiT 120 | DiT 224, Qwen3.5 123, optim 26 ms

1 · fp32 action head	201 ms fp32 GEMMs vs 66 ms bf16 — 2.96B DiT outside autocast	**bf16 DiT (−218 ms)**
2 · CPU inside forward()	first ~40 ms of every step: compute idle, HF preprocessing on critical path	**collate in DataLoader workers**
3 · Per-step syncs	loss readback, DS timers, MRoPE position-ids, mm-merge check	**sync-free logging + caches (2+3: −117)**
4 · Launch-bound eager	13,868 launches @ 8 µs median; occupancy fwd 34% vs bwd 70%; idle 205 ms	**CUDA Graphs + compile (−126 ms)**
5 · Comm configuration	15 small AR buckets; 22 ms redundant flatten; exposed AllGather tail	**bucket 1.5e9 + reduce_scatter off (−22)**

Optimization Roadmap

Actual step time (ms) after each optimization lands — same node, steady-state p50

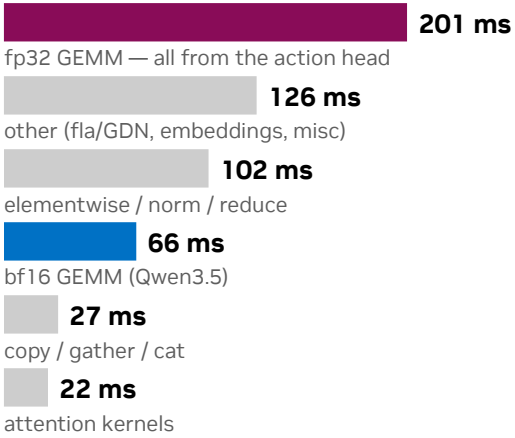


Cumulative levels re-measured per step with nsys attached on rank 0 (~5% overhead). The torch 2.7.1 + NCCL 2.26.2 environment upgrade landed between the bf16 and pipeline levels and is folded into the Pipeline bar. Final 249 ms = clean steady-state p50 of merged main.

Fix 1 · bf16 Action Head: −218 ms

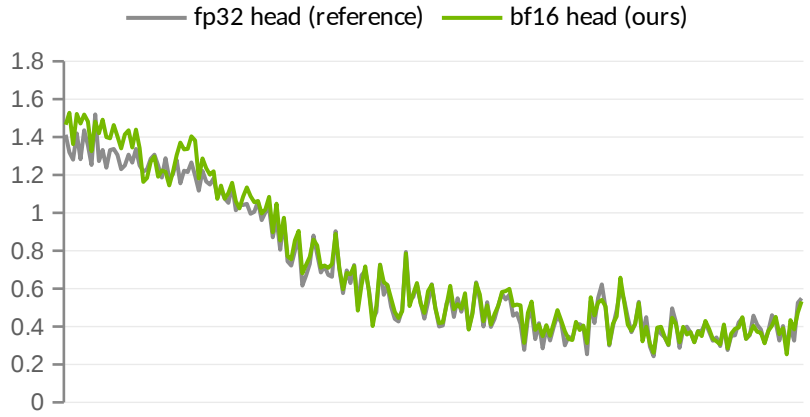
800 → 582 ms. One config knob — and numerically a no-op for training

GPU busy by kernel class (baseline step, 543 ms)



37% of all GPU math is fp32 — all of it from the 2.96B DiT outside the bf16 autocast region.

Same seed, 200 steps: loss is unchanged

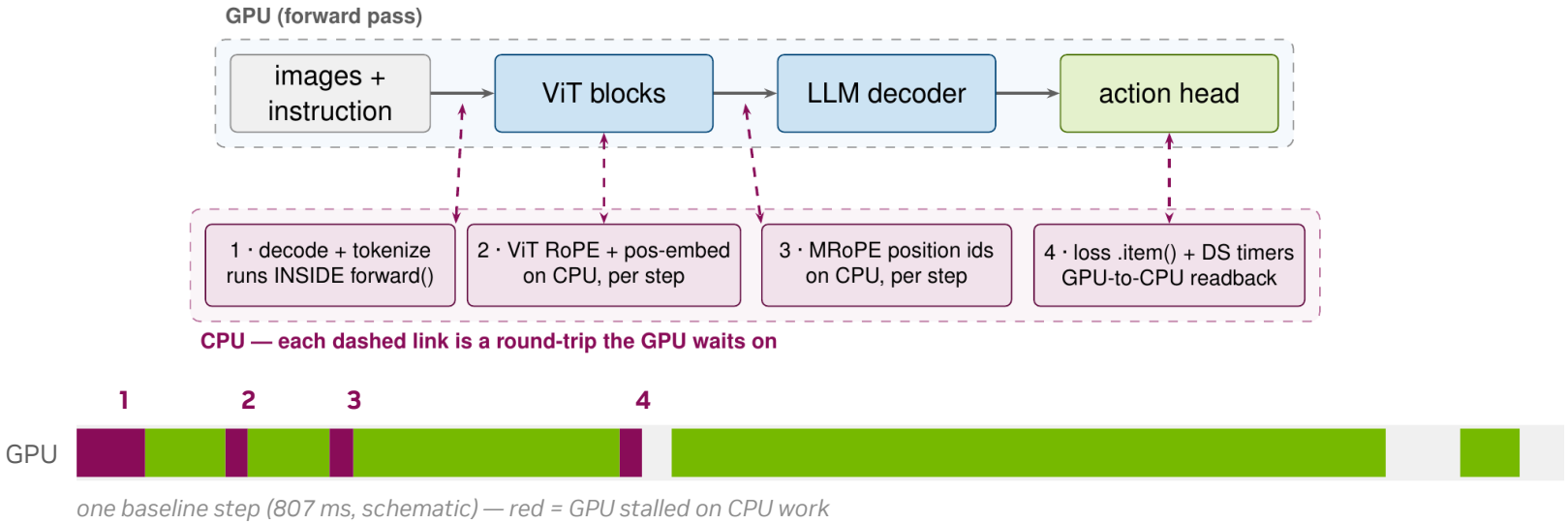


action loss, training steps 1 → 200 · mean per-step |Δ| = 0.04 (run-to-run noise floor) · final: 0.549 vs 0.533

Why was it fp32 at all? Inherited caution for diffusion numerics. Only the timestep embedding and the loss math need fp32 — we kept exactly those in fp32 and moved the transformer body to bf16 (dit_dtype knob, one-line rollback).

Sync Points: Where the CPU Blocks the GPU

Four CPU round-trips hide inside every baseline training step



one baseline step (807 ms, schematic) — red = GPU stalled on CPU work

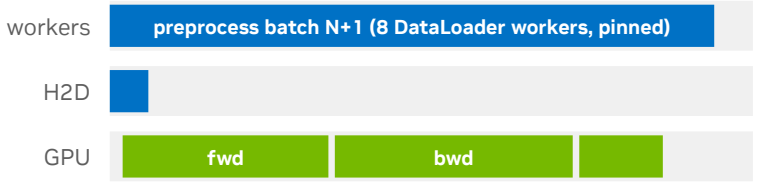
Measured: stall 1 ≈ 37 ms at every step start; stall 3 ≈ 13 ms; stall 4 drains the launch pipeline twice per step — together seeding the 205 ms/step of GPU idle. The next page removes all four.

Removing the Sync Points: −117 ms

545 → 428 ms. Every CPU round-trip moves off the critical path

Before — everything serialized inside forward()

After — CPU work overlaps the previous step



red = per-step syncs (loss readback, DS timers, mm-merge check): each one drains the launch pipeline and strands the GPU

Preprocess in workers	HF processor (image patches + tokenize) runs in 8 workers, overlapped with the previous step	**~40 ms off the critical path**
Side-stream H2D	pinned host batch + dedicated copy stream, overlaps the ZeRO-2 optimizer tail	**upload hidden**
Sync-free logging	DeepSpeed timers unsynchronized; loss .item() only on logging steps	**2 device syncs / step removed**
MRoPE position-id cache	3D position ids precomputed on CPU, cached by image grid	**~13 ms sync eliminated**
+ Mixed attention	vision tower FA2 varlen (16 images / kernel); text stays SDPA — full FA2 measured slower at seq 192	**−30 ms (428 → 398)**

Why CUDA Graphs: Launch-Bound Forward

Anatomy of the Qwen3.5 forward — same math, very different wall clock

Baseline (eager)

148 ms

3,929 kernel launches · 19.1 ms of cudaLaunchKernel CPU time · occupancy 34%

Optimized (CUDA Graphs)

65 ms

6 graph launches + 736 eager · 9.4 ms launch CPU · occupancy 57%

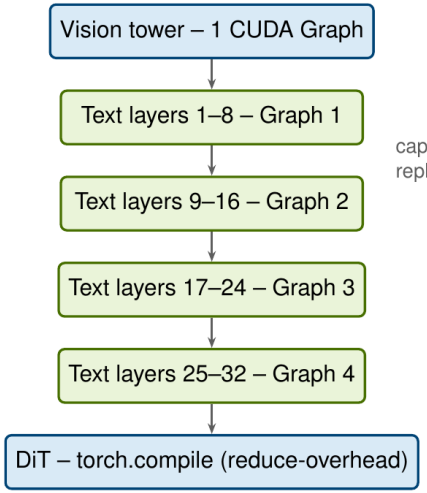
2.3× faster forward wall clock — with 81% fewer launches

- An eager op costs 30–60 µs of Python dispatch + launch; the median kernel runs 8 µs — the GPU finishes before the CPU can feed the next one
- Backward suffers far less (C++ autograd engine, occupancy 70%) — forward is where launch overhead bites
- A CUDA Graph replays an entire 8-layer group as ONE cudaGraphLaunch — per-op dispatch leaves the critical path

Busy also shrinks 50 → 37 ms (mixed attention + fusion); the split isolates the launch effect. Windows: baseline Qwen3.5 fwd vs merged-main VLM fwd, same launch-window accounting.

CUDA Graphs — and Why Four, Not One

398 → 271 ms (−126). Capture once, replay every step



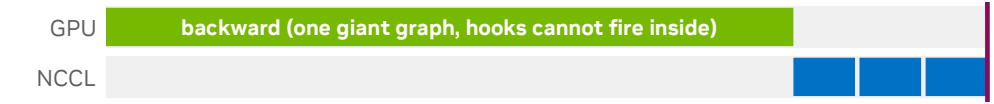
Enablers: static shapes (pad-192) · vision tower = 1 graph · DiT = torch.compile · merge sync-free eager

Why is the decoder FOUR graphs, not one?

DeepSpeed's compute-comm overlap must survive: AllReduce hooks fire only BETWEEN replays — never inside one.

4 graphs — AllReduce overlaps the group replays:

1 graph — every AllReduce waits for the whole backward:



same compute, same bytes — the red line is where the step ends

Kernel launches: 13.9k → 6.4k per step. CPU launch cost leaves the critical path; GPU idle collapses.

ZeRO-2 Comm Tuning: Two Knobs

271 → 249 ms. Two different problems: a redundant flatten, and a blocked backward

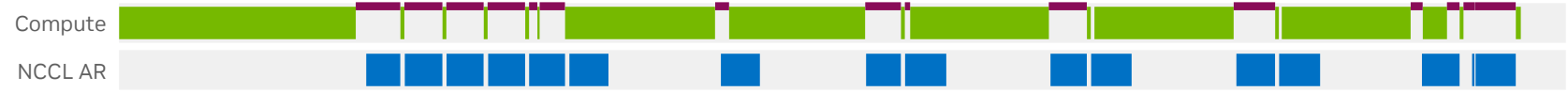
Knob 1 · reduce_scatter → false (−13 ms)

- Default path flattens every bucket (22 ms/step CatArrayBatchedCopy) — but grads already sit contiguous
- Plain AllReduce skips it. Cost: ~2× wire bytes — fine while comm overlaps and NVLink has headroom

Knob 2 · reduce_bucket_size → 1.5e9 (−12 ms)

- Grads double-buffer into TWO buckets; every boundary makes compute wait on the reduction stream
- Too small → backward stalls behind its own AllReduces (below). Too big → exposed tail (2e9 worse)

bucket 5e8 — backward 167 ms · 18 AllReduces · 15 stalls, 47 ms of blocked compute



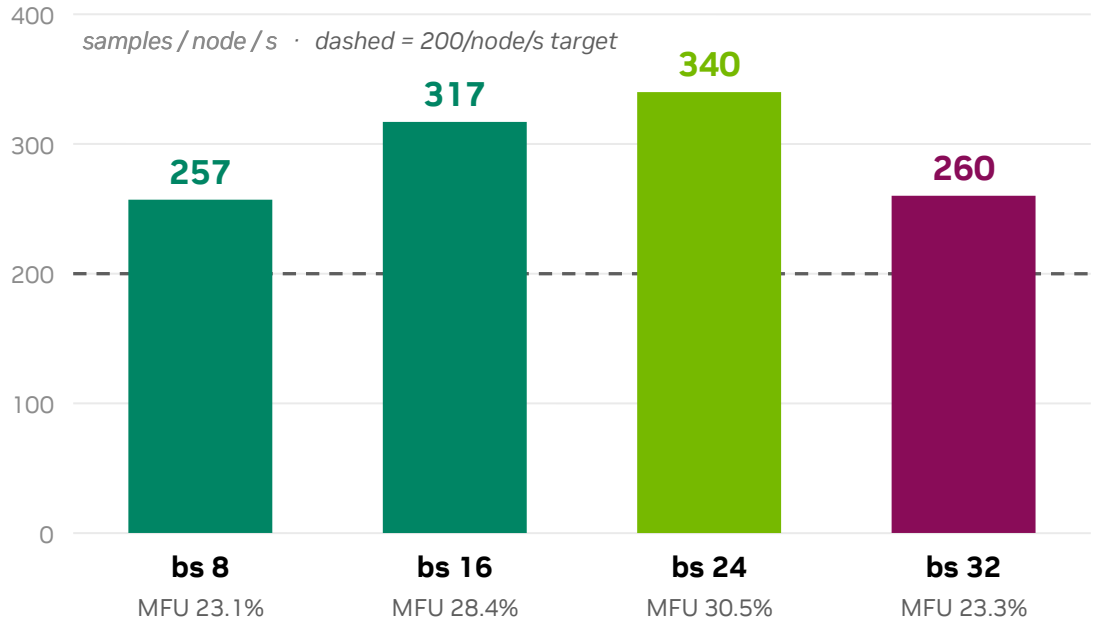
bucket 1.5e9 — backward 154 ms · 8 AllReduces · 8 stalls, 27 ms

one backward pass, real kernels (node-level nsys, same node, same bytes) · red = compute stream stalled at a bucket rendezvous

No conflict — they compose: bucket size = how OFTEN compute and reduction rendezvous; reduce_scatter = what each rendezvous COSTS. false makes each AR bigger, strengthening the case for large buckets.

Batch Scaling: Sweet Spot at 24 per GPU

Same optimized config, only per-GPU batch changes — peak 340 samples/node/s



- Batching amortizes launch and comm costs: MFU climbs 23% → 30.5%
- bs 32 regresses hard (986 ms/step): not OOM, not the dataloader — memory-pressure effects under investigation
- Larger global batch (192) changes optimization dynamics — needs a convergence run before adoption

What's Next — and What Not to Retry

Levers ranked by ceiling — and measured dead ends

Next levers (ranked by ceiling)

fp8 for the backward GEMMs

Sustained GEMM throughput caps at ~622 TFLOPS on this node (power wall, not occupancy) — fp8 is the only math lever left.

Gradient compression / 1-bit Adam

NVLink is at its ceiling, so bytes are the only comm lever for the ~30 ms of exposed AllReduce. Needs convergence sign-off.

Overlap the param AllGather (44 ms)

The step-end AllGather is now fully exposed and ZeRO-2 cannot overlap it — adopt FSDP, or add an async gather to ZeRO-2.

Deployment & batch

Split-topology nodes (4+4 NVLink) need NCCL_P2P_LEVEL=NVL. bs 24 (340 samples/node/s) is gated on a convergence run.

Don't retry — we measured

FA2 for the text stack too

+10 ms at seq 192: unpad/repad overhead outweighs the kernel gains. Mixed (vision-only FA2) is the right split.

NCCL knob tuning (NVLS / protocol / channels)

AllReduce microbench already hits 478 GB/s busbw — the NVLink per-direction ceiling. Config cannot add bandwidth.

One giant decoder graph

Library graph-breaks fragment the capture, and a single graph would defer every AllReduce to the step end (page 10).

FlashQLA GDN kernels

Total GDN kernel time is only ~10.5 ms/step at seq 192 — below the integration risk; also needs torch $\geq$ 2.8.